

Module 16 – Cloud Fundamentals

1. Introduction to Cloud Computing

What is Cloud Computing?

Cloud Computing is the delivery of computing resources such as servers, storage, databases, networking, software, and analytics over the internet on a **pay-as-you-go** basis. Instead of maintaining physical hardware, organizations can rent resources from cloud providers whenever needed.

Advantages of Cloud Computing

- Reduced infrastructure cost.
- High availability.
- Easy scalability.
- Global accessibility.
- Automatic updates.
- Faster deployment.

Challenges of Cloud Computing

- Internet dependency.
- Data privacy concerns.
- Vendor lock-in.
- Compliance requirements.

2. Virtualization

Virtualization is the process of creating virtual versions of physical resources such as servers, storage, and operating systems. A single physical server can run multiple Virtual Machines (VMs) using a **Hypervisor**.

Benefits

- Better hardware utilization.
- Lower infrastructure cost.
- Faster server provisioning.
- Easier backup and recovery.

3. Service-Oriented Architecture (SOA)

SOA is a software design approach where applications are built as independent services that communicate over a network.

Characteristics

- Loosely coupled services.
- Reusable components.
- Independent deployment.
- Standard communication (REST, SOAP).

4. Cloud vs On-Premises

Cloud Computing	On-Premises
Resources rented	Own physical infrastructure
Low initial cost	High setup cost
Automatic scaling	Manual scaling
Managed by provider	Managed by organization
Faster deployment	Slower deployment

5. Cloud Service Models

Infrastructure as a Service (IaaS)

Provides virtual machines, storage, and networking.

Examples: Amazon EC2, Azure Virtual Machines.

Platform as a Service (PaaS)

Provides a platform for application development without managing infrastructure.

Examples: AWS Elastic Beanstalk, Google App Engine.

Software as a Service (SaaS)

Provides ready-to-use software over the internet.

Examples: Gmail, Microsoft 365, Salesforce.

6. Cloud Deployment Models

Public Cloud

Cloud resources are shared among multiple users.

Example: AWS, Azure, GCP.

Private Cloud

Cloud infrastructure is dedicated to a single organization.

Hybrid Cloud

Combination of Public and Private Cloud.

Community Cloud

Shared by organizations with similar requirements.

7. Cloud Service Providers

Provider	Popular Services
AWS	EC2, S3, RDS, Lambda
Microsoft Azure	Virtual Machines, SQL Database
Google Cloud Platform (GCP)	Compute Engine, BigQuery

8. Amazon EC2 (Elastic Compute Cloud)

Amazon EC2 provides scalable virtual servers called **Instances**.

Features

- On-demand virtual servers.
- Easy scaling.
- Pay-as-you-go pricing.
- Multiple instance types.

EC2 Components

- Instance
- AMI (Amazon Machine Image)
- Security Groups
- Key Pair

9. Amazon Machine Image (AMI)

An AMI is a preconfigured template used to create EC2 instances.

Types

- AWS AMI
- Community AMI
- Marketplace AMI
- Custom AMI

10. Security Groups

Security Groups act as a **virtual firewall** for EC2 instances.

Common Ports:

- **22** – SSH
- **80** – HTTP
- **443** – HTTPS

11. Key Pair

A Key Pair is used for secure login to EC2 instances.

It consists of:

- Public Key
- Private Key (.pem)

12. Amazon ECS

Amazon ECS (Elastic Container Service) is a fully managed container orchestration service used to deploy and manage Docker containers.

ECS	EC2
Manages containers	Provides virtual machines

Uses Docker	Runs operating systems
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13. Amazon S3

Amazon S3 (Simple Storage Service) is an object storage service.

Component

- Bucket
- Object

Storage Classes

- Standard
- Intelligent-Tiering
- Standard-IA
- Glacier

Features

- Versioning
- Lifecycle Policies
- High Durability

14. Amazon EBS

Amazon EBS (Elastic Block Store) provides persistent block storage for EC2 instances.

Volume Types

- gp2
- gp3
- io1

Features

- Persistent storage.
- Snapshots.
- High performance.

15. Amazon VPC

Amazon VPC (Virtual Private Cloud) is an isolated virtual network in AWS.

Components

- Public Subnet
- Private Subnet
- Route Table
- Internet Gateway
- NAT Gateway
- Security Groups

16. Elastic Load Balancer (ELB)

ELB distributes incoming traffic across multiple servers.

Application Load Balancer (ALB)

- Layer 7
- HTTP/HTTPS
- Path-based routing

Network Load Balancer (NLB)

- Layer 4
- TCP/UDP
- High performance

17. Amazon RDS

Amazon RDS is a managed relational database service.

Supported Databases

- MySQL
- PostgreSQL
- SQL Server
- Oracle
- Amazon Aurora

Features

- Automated backups.
- Multi-AZ deployment.
- Automatic failover.

18. Amazon DynamoDB

Amazon DynamoDB is a fully managed NoSQL database.

Features

- Key-value database.
- Automatic scaling.
- Low latency.
- High availability.

19. AWS Lambda

AWS Lambda is a **serverless computing service** that executes code in response to events without managing servers.

Features

- Event-driven.
- Automatic scaling.
- Pay-per-execution.
- Supports Java, Python, Node.js.

20. AWS API Gateway

AWS API Gateway is used to create, publish, secure, and manage REST APIs.

Features

- REST API creation.
- Lambda integration.
- Deployment stages (Dev, Staging, Production).
- Authentication.
- Throttling.
- HTTPS security.